U4RASD at KnowledgeGraphEval-2026: Ontology-Constrained Encoder and LLM Systems for Arabic Relation Extraction

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Abstract

This paper details our submission to KnowledgeGraphEval 2026 Subtask 2 on directed Arabic relation extraction. We explored encoder-based and large language model approaches for predicting one of 40 relations or *no_relation* from an Arabic sentence and a subject–object pair. Our encoder systems use typed entity markers and separate relation-existence and relation-classification heads, while our LLM systems fine-tune Yehia and Qwen3.5 with QLoRA and ontology-constrained prediction. Our best system, an ensemble of Yehia-RE, Qwen-RE, and TypedPair, achieved a micro-F1 score of 94.06 and ranked **1st** in Subtask 2. Yehia-RE (full-vocabulary) achieved 92.98, while the best encoder and Qwen-RE achieved 92.69 and 91.94 respectively. Our results highlight the importance of constrained prediction and complementary modeling approaches for Arabic relation extraction.

1 Introduction

Relation Extraction (RE) is an information extraction task that aims to identify semantic relations between entities mentioned in text. In directed RE, the relation is predicted from a designated subject to a designated object. Arabic RE remains challenging when relations are semantically similar, entity mentions occur multiple times in the same sentence, or entity-type information is needed to determine which relations are valid.

In this paper, we present our submission to KnowledgeGraphEval 2026 Subtask 2 on directed Arabic relation extraction. Given an Arabic sentence and a subject–object pair, the task is to predict one of 40 positive relations or *no_relation* when no valid relation is expressed (Aljabari et al., 2026). The task data are also highly imbalanced, making accurate positive-relation prediction and *no_relation* handling particularly important.

We investigate two system families. The first, TypedPair, uses Arabic encoder models with typed directional entity markers and separate heads for relation existence and positive-relation classification. The second uses Yehia (Navid-AI, 2025) and Qwen3.5 (Qwen Team, 2026), adapted with QLoRA (Dettmers et al., 2023), resulting in our Yehia-RE and Qwen-RE systems. Yehia-RE formulates relation prediction as a constrained Arabic one-token multiple-choice task, while Qwen-RE is evaluated using both causal-LM prediction and sequence classification. All systems share entity-occurrence resolution, entity-type prediction, and ontology-based relation filtering.

Our main contributions are: (1) the TypedPair encoder family; (2) ontology-constrained Yehia-RE and Qwen-RE systems; and (3) a shared pipeline for occurrence resolution, entity typing, and relation filtering. Our best ensemble achieved a micro-F_1 score of 94.06 and ranked **1st** in Subtask 2. The strongest individual Yehia-RE, TypedPair, and Qwen-RE systems achieved 92.98, 92.69, and 91.94, respectively. The trained models and released resources are available at `https://huggingface.co/U4RASD`.

2 Background

The task data are derived from Wojood^{Relations} (Aljabari et al., 2025b), while WojoodOntology provides relation definitions and domain–range constraints over entity types (Aljabari et al., 2025a). Konooz (Hamad et al., 2025) is used in the shared task's NER subtask; our submission addresses only Subtask 2 on relation extraction.

Prior relation-extraction work has explored marker-based encoder models, including entity markers (Baldini Soares et al., 2019), PL-Marker (Ye et al., 2022), and typed entity markers (Zhou and Chen, 2022). For Arabic RE, Wojood^{Relations} introduces NLI-RE, which formulates candidate

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relations as natural-language-inference hypotheses and filters them according to compatible entity types (Aljabari et al., 2025b).

Generative approaches have also modeled RE through sequence generation and prompting. AREEj augments Arabic relation extraction with textual evidence supervision (Rakan Al Mraikhat et al., 2024), while GPT-RE and related work explore in-context and instruction-based relation prediction (Wan et al., 2023; Wadhwa et al., 2023; Zhang et al., 2023). Our work investigates both encoder-based and constrained LLM-based approaches under the same ontology-aware Arabic RE setting.

3 System Overview

Figure 1 summarizes the shared full pipeline and the two model families. The following subsections summarize the components that distinguish the submitted systems.

3.1 Shared Pipeline

The input provides the subject and object strings, but repeated occurrences may require resolving which mentions should be evaluated, and entity types are not directly provided to the relation models. We therefore locate all exact subject and object occurrences and construct the possible directed occurrence pairs. Each occurrence is then assigned an entity type using our TypePredictor, and the predicted types are used both in the typed entity markers and to restrict the candidate relations according to WojoodOntology. When multiple occurrence pairs are possible, the system applies a row-level decision strategy to select or combine their predictions. TypePredictor and candidate-pair handling are detailed in Appendices B and C.

3.2 Encoder-Based Systems

The TypedPair family inserts typed directional markers around the selected subject and object occurrences. It combines the sentence-start state, the two opening-marker states, simple pair interactions, and a 42-dimensional subject–object type vector. A multilayer perceptron feeds two heads: a binary head predicts whether a positive relation exists, and a 40-way head selects the positive relation. Negative rows supervise only the existence head, avoiding an arbitrary positive-class target. At inference, the positive head is masked by the predicted type pair and is consulted only when the existence score passes a validation-selected threshold.

We evaluate NeoAraBERT (Abou Chakra et al., 2026), AraBERTv02 (Antoun et al., 2020), ARBERTv2 (Elmadany et al., 2023), AraBERTv02-Large (Antoun et al., 2020), and AraModernBERT (Elshehy et al., 2026) backbones, together with various row-level candidate-pair decision strategies. The exact representation, loss, threshold selection, tensor dimensions, and candidate algorithms are provided in Appendix D.

3.3 LLM-Based Systems

3.3.1 Yehia-RE

We fine-tune Yehia with QLoRA and formulate each relation decision as a row-specific Arabic multiple-choice problem. The marked sentence, predicted entity types, and only the ontology-compatible positive relations are shown in the prompt, followed by *no_relation*. Each displayed option is mapped to an existing Arabic one-token code, and the model returns the code alone.

We evaluate two Yehia-RE training variants. The full-vocabulary variant optimizes the gold code against the complete model vocabulary and, at inference, compares only the displayed option codes. The row-local variant instead normalizes the training objective only over the option codes displayed for the current example. Full objectives and candidate-pair aggregation are detailed in Appendix E.

3.3.2 Qwen-RE

We fine-tune Qwen3.5-4B and Qwen3.5-9B with QLoRA using prompts containing the marked Arabic sentence, entity strings and types, and concise meanings for the ontology-compatible labels. We denote the causal-LM variants as Qwen-RE-4B and Qwen-RE-9B, and the 9B sequence-classification variant as Qwen-RE-9B-Cls. The markers identify the entity occurrences in the sentence and make the subject–object direction explicit. The causal-LM systems are trained to produce the relation label. At inference, we either generate the label or append each compatible label and score it using its mean token log-probability. Because the output label set is fixed, we additionally fine-tune Qwen3.5-9B with a trainable 41-way sequence-classification head. This classifier receives a shorter prompt and directly predicts a relation class, with ontology-incompatible classes masked during both training and inference, avoiding textual generation and parsing failures.

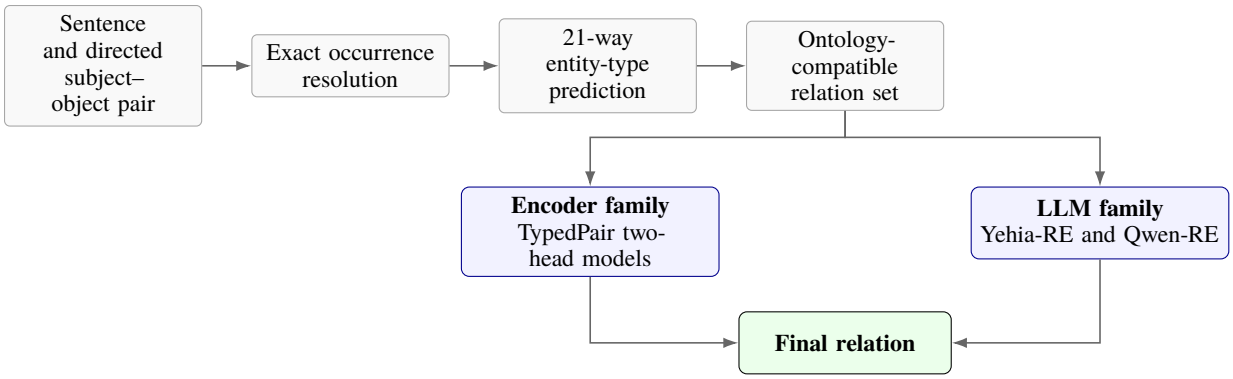


Figure 1: Shared relation-extraction pipeline and the two model families.

The Qwen prompt structures, a paired causal-LM/classifier example, complete-label scoring, class-imbalance interventions, and training configuration appear in Appendix F. We also tested manual disambiguation rules and GEPA prompt evolution (Agrawal et al., 2026); these exploratory interventions did not replace the selected submitted configurations and are documented separately in Appendix G.

4 Experimental Setup

4.1 Data and Preprocessing

The labeled training data were enriched with occurrence-specific character spans and entity types. We first aligned relation mentions with the Wojood flat and nested NER annotations (Jarrar et al., 2022); unresolved mentions were then matched against predictions from the Wojood NER model. Two authors manually resolved the remaining 240 training rows that could not be enriched automatically, retaining the original relation labels. This enrichment was used only for training; blind-test rows were processed entirely by the full inference pipeline. Conflict filtering, ontology compatibility filtering, and deduplication reduced 17,381 raw relation rows to 15,686 training instances; the development artifact contains 1,687 rows, while the official blind evaluation input contains 2,074 rows. The development artifact contains both positive relations and *no_relation* examples, whereas the official blind evaluation input contains only positive-relation rows; the resulting difference in class distribution means development and blind micro-F_1 scores are not directly comparable. Complete processing details are given in Appendix A.

The processed training distribution is strongly imbalanced: *no_relation* has 6,677 examples, followed by `Location.located_in` with 4,269 and `PartOf.geopolitical_division` with 1,684. Encoder models use cross-entropy objectives, Yehia-RE uses frequency-based sampling, the Qwen-RE causal-LM experiments use clipped inverse-frequency weighting, and the Qwen-RE classifier uses square-root inverse-frequency sampling. Selected relation-frequency statistics are reported in Appendix A, while model-specific sampling and weighting details are reported in Appendices D, E, and F.

4.2 Training, Decoding, and Selection

A cross-system summary of the principal training and selection configurations is provided in Table 21 in Appendix I. TypePredictor and TypedPair encoders are fully fine-tuned. Yehia and Qwen use 4-bit NF4 QLoRA with frozen base weights. Marker handling after tokenization follows the procedure described in Appendix C. Relation-extraction checkpoints are selected using the 1,687-row development artifact and then frozen before official inference. Fixed runner settings are reported separately in the corresponding system appendices. TypePredictor is selected independently using its entity-typing validation split.

Efficiency. AraBERTv02-Large TypedPair required approximately 43 minutes on a single NVIDIA A40, while Yehia-RE required approximately 3.32 hours on a single NVIDIA L40S. Qwen-RE-9B required approximately 12 hours on a single NVIDIA RTX PRO 4500. Since these measurements were obtained on different GPUs, they should not be treated as a controlled efficiency comparison, but they illustrate the substantially higher training cost of the LLM-based systems relative to the encoder-based model.

Official evaluation uses micro-F_1. Depending on the configuration, the row-level deci-

sion uses either majority voting, top-existence/top-confidence selection, max-positive selection, or nearest-candidate selection. Detailed definitions and deterministic tie-breaking are provided in Appendices C, D, E, and F.

5 Results

Table 1 summarizes representative official submissions. All values are full-pipeline scores on the blind set; post-submission prompt analyses are reported separately in Appendix G. The shared task permitted up to 30 official submissions per team; we used 22.

5.1 Encoder Results

Across the official blind submissions, AraModernBERT achieved the strongest encoder result at 92.69, compared with 91.96 for ARBERTv2 and 92.20–92.29 for the two reported AraBERTv02-Large aggregation strategies. NeoAraBERT with majority + logit-sum obtained 88.33. A component ablation on AraModernBERT (Appendix D.6) shows that the complete configuration with both typed text markers and the directional type vector achieved the strongest performance among the tested variants. The complete encoder and aggregation results are reported in Appendix D.5.

5.2 LLM Results

Yehia-RE (full-vocabulary) achieved 92.98, while the row-local variant reached 92.84 with max-positive aggregation. Its constrained one-token formulation eliminates free-form label formatting errors by casting relation prediction as a row-specific Arabic multiple-choice task. Our best system, a three-system majority ensemble of Yehia-RE, Qwen-RE, and TypedPair, achieved 94.06 and ranked **1st** in Subtask 2. Detailed Yehia-RE formulations, candidate aggregation, and official results are provided in Appendix E.

On the official blind evaluation, Qwen-RE-4B and Qwen-RE-9B obtained 91.53 and 91.76, respectively, while Qwen-RE-9B-Cls obtained 91.94. These blind-set differences are reported descriptively rather than as controlled ablations. Complete official Qwen results are reported in Appendix F.8.

5.3 Error Analysis

The main failure modes reflect interactions among pipeline components. A type error can change the typed markers and remove the correct relation from the compatible set. Repeated surface forms can associate the prediction with the wrong textual evidence. Semantically close relations, especially generic location versus headquarters or administrative division, are difficult under a highly skewed label distribution. Finally, causal-generation systems may understand the relation but emit an invalid or partial canonical label. Additional qualitative error analysis is provided in Appendix H, while the exploratory disambiguation experiments are reported in Appendix G.

6 Conclusion

We presented two complementary system families for KnowledgeGraphEval 2026 Arabic relation extraction. The encoder family combines typed directional markers with a two-head relation objective, while the LLM family constrains Yehia-RE and Qwen-RE to ontology-compatible decisions. Our best system, an ensemble of Yehia-RE, Qwen-RE, and TypedPair, achieved an official micro-F_1 score of 94.06 and ranked **1st** in Subtask 2. Yehia-RE (full-vocabulary) achieved 92.98, while the strongest encoder and Qwen-RE systems reached 92.69 and 91.94, respectively. The results highlight the importance of constrained prediction, entity-type compatibility, candidate resolution, and complementary modeling approaches. Future work will propagate type uncertainty, jointly score occurrence and relation hypotheses, and study confidence-weighted ensembles.

Limitations

The reported training-level results are based on a single random seed. Consequently, small differences between model configurations should be interpreted cautiously, as we cannot quantify variation across training runs. The systems also depend on hard entity-type predictions and exact-match occurrence recovery, and they remain challenging for rare or semantically adjacent relations. Because the TypePredictor dataset uses a mention-level split, sentence-level overlap with relation-extraction inputs is possible. Although TypePredictor never uses relation labels, such overlap may affect the entity-type predictions used for typed markers and ontology filtering.

System	Training/decoder	Row-level decision	Official F_1
TypedPair–NeoAraBERT	Two-head encoder	Majority + logit-sum	88.33
TypedPair–ARBERTv2	Two-head encoder	Majority + logit-sum	91.96
TypedPair–AraBERTv02-Large	Two-head encoder	Top existence	92.29
TypedPair–AraModernBERT	Two-head encoder	Majority + logit-sum	**92.69**
Yehia-RE (row-local)	Row-local CE	Max positive	92.84
Yehia-RE (full-vocabulary)	Full-vocabulary CE	Max positive	**92.98**
Qwen-RE-4B	Causal-LM label generation	Nearest candidate	91.53
Qwen-RE-9B	Causal-LM label generation	Nearest candidate	91.76
Qwen-RE-9B-Cls	Sequence classification	Nearest candidate	**91.94**
Team ensemble	Yehia-RE (full-vocabulary) + Qwen-RE-9B-Cls + TypedPair–AraModernBERT	Majority vote; Yehia-RE tie-break	**94.06**

Table 1: Summary of representative official KnowledgeGraphEval 2026 Subtask 2 submissions. Bold values mark the strongest configuration within each model family.

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A Data Processing and Artifacts

A.1 Source Resources and Processed Schema

The shared-task data are derived from Wojood^{Relations}, which extends Wojood with 40 directed relation types (Aljabari et al., 2025b). Wojood supplies nested entity annotations over 21 types (Jarrar et al., 2022), while WojoodOntology supplies relation definitions and directed domain–range constraints (Aljabari et al., 2025a).

We retained the original sentence, subject, object, and relation fields unchanged and added occurrence-specific character offsets, token offsets, and entity types. These additions support typed marker insertion, span validation, and ontology compatibility checks.

A.2 Verified Raw and Enriched Example

Table 2 shows a verified row from the original Wojood^{Relations} training file. The row illustrates the task input before our span and type enrichment.

The corresponding processed row retains all original fields and adds the resolved occurrence information shown in Table 3. This example is particularly useful because the object occurs inside the subject span, illustrating a nested mention case.

A.3 Artifacts and Processing Stages

The blind input contained no gold relation labels. It was used only to produce official predictions; we did not manually inspect, infer, or reconstruct its hidden labels.

A.4 Span and Entity-Type Recovery

Training relation rows provide entity surface forms but not the occurrence-specific spans and types required by our systems. We first align mentions with Wojood flat and nested NER annotations (Jarrar et al., 2022), preferring exact span matches and allowing compatible containment matches when necessary. Unresolved mentions are then matched against predictions from the Wojood NER model. After these automatic steps, two authors manually resolved the spans and entity types of the remaining 240 training rows, while retaining their original relation labels.

No manual span or type annotation is used for the blind set; its occurrences and entity types are determined entirely by the inference pipeline described in Appendix C.

A.5 Span Validation

For every processed row, we verified that the character slice identified by the subject offsets reproduced the subject string and that the object slice reproduced the object string. Rows with mismatched offsets were rejected rather than silently accepted. The final processed training and development artifacts reported zero slice errors.

This check is important because selecting the wrong repeated occurrence can associate a relation label with unrelated textual evidence. At blind inference, the systems resolve possible occurrence pairs as described in Appendix C.

A.6 Conflict Removal and Deduplication

We first removed groups containing contradictory annotations for the same resolved sentence and entity pair, reducing 17,381 raw rows to 17,208 non-conflicting rows. We then removed exact duplicates sharing the sentence, subject, object, and relation label. Together with the ontology-compatibility filtering described below, these preprocessing steps produced 15,686 final training instances.

A.7 Ontology Compatibility and Operational Mapping

For each positive training instance, we checked the resolved subject and object types against the directed domain–range constraints defined by WojoodOntology (Aljabari et al., 2025a). Because the task is directional, compatibility is evaluated specifically for the ordered pair (τ_s, τ_o). Training rows whose subject–object type pair was not a valid domain–range combination for the labeled relation were removed during preprocessing. This prevents the models from being trained on relation instances that are incompatible with the ontology under the resolved entity types.

We made one systematic modification to the operational compatibility map based on a highly frequent pattern observed in the labeled training data. In the original WojoodOntology constraints, the domain and range of `Location.located_in` are more restricted. However, the relation data repeatedly contain labeled `Location.located_in` examples involving ORG, FAC, LOC, and GPE entities in both subject and object positions. We therefore expanded the operational domain–range set for this relation to

$$\begin{aligned} r &= \texttt{Location.located_in}, \\ \text{Domain}(r) &= \{\text{ORG}, \text{FAC}, \text{LOC}, \text{GPE}\}, \\ \text{Range}(r) &= \{\text{ORG}, \text{FAC}, \text{LOC}, \text{GPE}\}. \end{aligned}$$

This modification was motivated by the recurring annotation pattern in the provided relation data and was applied consistently during preprocessing and ontology-based candidate filtering. It is an implementation-level adaptation used by our system; it does not alter the published WojoodOntol-

Field	Stored value
Sentence ID	`100002243`
Triple ID	`3830`
Sentence	بلدية قلقيلية : الى قائمقام طولكرم / مدير منطقة قلقيلية : قرارات مجلس محلي قلقيلية : بتاريخ (2 / 1 / 1964) .
Subject	بلدية قلقيلية
Object	قلقيلية
Relation	`Location.located_in`

Table 2: Verified example from the original Wojood[Relations] training artifact before our enrichment.

Entity	Subject	Object
Surface form	بلدية قلقيلية	قلقيلية
Character span	$(0, 13)$	$(6, 13)$
Token span	$(0, 1)$	$(1, 1)$
Entity type	`ORG`	`GPE`

Table 3: Occurrence-specific information added to the verified relation row in Table 2. Offsets are reproduced as stored in the processed artifact.

Processing stage	Rows
Raw relation rows	17,381
After conflict removal	17,208
After ontology alignment and deduplication	15,686
Development artifact	1,687
Blind official input	2,074

Table 4: Relation artifact sizes and preprocessing stages.

Relation	Training rows
no_relation	6,677
`Location.located_in`	4,269
`PartOf.geopolitical_division`	1,684
`Location.lives_in`	876
`Personal.has_occupation`	620
`Location.headquartered_in`	172

Table 5: Selected frequent and difficult relations in the processed training artifact.

ogy itself or the relation labels provided by the shared task.

A.8 Relation and Entity-Type Distributions

The subject side is dominated by organizations, persons, and geopolitical entities, while the object side is dominated by geopolitical entities, organizations, occupations, and locations. Frequent directed type pairs therefore include organization–place, place–place, organization–organization, person–place, and person–occupation.

B TypePredictor

B.1 Dataset Construction

The TypePrediction collection was curated from the Wojood and WojoodFine NER resources (Jarrar et al., 2022; Liqreina et al., 2023). Each example contains an Arabic sentence, one exact mention occurrence, and its entity type. TypePredictor is trained only on entity-type labels and never uses relation labels. Relation-extraction rows were excluded from this curation. Exact duplicate mentions were collapsed, and same-span examples with conflicting labels were removed.

The resulting collection contains 125,996 examples over 21 types: 100,796 for training, 12,600 for validation, and 12,600 for testing. The split is a shuffled, stratified, mention-level 80/10/10 split. It was produced using two stratified `train_test_split` operations with `random_state=42`. The same sentence may occur in multiple splits with different entity spans. Because the split is mention-level, sentence-level overlap with relation-extraction inputs, including the blind evaluation input, is possible. Such overlap may affect the entity-type predictions subsequently used for typed markers and ontology filtering, but it does not expose relation labels to TypePredictor. Validation and test examples are therefore additionally categorized as `seen_sentence_new_entity` or `unseen_sentence`.

GPE	ORG	DATE	PERS	NORP	ORDINAL	OCC	EVENT
26,902	26,593	19,762	10,712	10,217	7,807	7,696	3,769
CARDINAL	LOC	WEBSITE	FAC	LAW	TIME	MONEY	CURR
3,724	2,388	1,478	1,296	904	871	420	411
LANGUAGE	PERCENT	PRODUCT	QUANTITY	UNIT			
332	312	190	106	106			

Table 6: TypePrediction mention counts over all splits.

B.2 Verified Example

Table 7 gives a verified TypePrediction training row. Unlike the relation artifacts, the TypePrediction files do not physically store the marked model input. The marked input is constructed at loading time from the sentence and stored character offsets.

Training rows store `entity`, `type`, `sentence`, `start_token`, `end_token`, `start_char`, `end_char`, `raw_types`, `sources`, and `id`; validation and test rows additionally store `evaluation_category`. The marked input itself is not stored.

For the example in Table 7, the generated input contains the fragment

كما تدير [ENT] المنظمة [/ENT] مركزاً للإصابات البالغة في مديرية الحوبان، وهو يعالج جرحى الحرب والمرضى الذين يحتاجون إلى الرعاية الطبية.

The model input is generated as

$$x' = x_{<s}\ \texttt{[ENT]}\ x_{s:e}\ \texttt{[/ENT]}\ x_{>e},$$

possibly within a selected context window. The tokens `[ENT]` and `[/ENT]` were explicitly added to the tokenizer as special tokens with IDs 65,000 and 65,001, respectively, and the encoder embedding matrix was resized to a vocabulary size of 65,002.

B.3 Architecture and Training

The TypePredictor consists of `U4RASD/NeoAraBERT` and a single classification layer. Let $H \in \mathbb{R}^{L\times 768}$ be the final hidden states and let $\mathbf{h}_{\text{cls}} = H_0$. The classifier computes

$$\begin{aligned}\mathbf{z}^{\text{type}} &= W_t\,\text{Dropout}_{0.1}(\mathbf{h}_{\text{cls}}) + \mathbf{b}_t,\\ \mathbf{z}^{\text{type}} &\in \mathbb{R}^{21}.\end{aligned} \tag{1}$$

The $768 \to 21$ head contains 16,149 parameters. Training minimizes unweighted multiclass cross-entropy and inference returns the highest-scoring type.

B.4 Detailed Evaluation

We further evaluate the TypePredictor using aggregate performance metrics, sentence-level generalization categories, per-type performance, and diagnostic visualizations. Overall validation and test performance is reported in Table 9, while Table 10 separates performance between seen sentences with new entity spans and entirely unseen sentences. Table 11 provides the detailed per-type precision, recall, and F_1 scores on the test split. Finally, Figure 2 presents the normalized confusion matrix, per-class F_1 scores, and performance across sentence categories.

C Shared Inference Pipeline

C.1 Marker Insertion and Overlap Handling

Markers are inserted from character spans rather than through global surface replacement. Insertions are ordered to keep nested, overlapping, adjacent, and equal-span cases deterministic. After tokenization, the system verifies that the required opening and closing marker IDs remain present. When necessary, the sentence is cropped around the outer entity boundaries.

C.2 Ontology Filtering

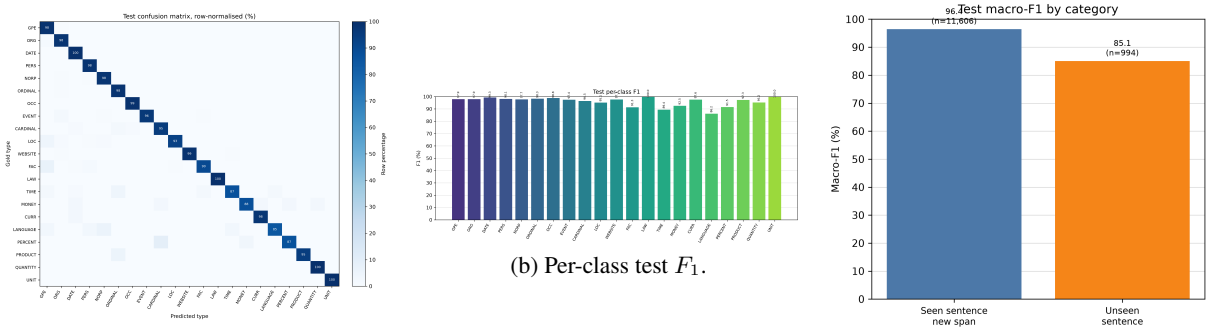
For a predicted directed type pair (τ_s, τ_o), the operational WojoodOntology map returns the compatible positive relations. *no_relation* remains available for every pair. Incompatible encoder and classifier logits are masked, while generative prompts display or score only compatible labels.

C.3 Candidate-Pair Handling

The model families use different strategies when several occurrence pairs are possible; there is no single nearest-distance rule shared by all released runners. TypedPair can score all candidate pairs and aggregate their predictions as described in Appendix D. Yehia-RE also scores candidate pairs and applies the row-level aggregation rules described in Appendix E.

Field	Stored value
ID (`id`)	`tp_093924`
Entity (`entity`)	المنظمة
Type (`type`)	`ORG`
Sentence (`sentence`)	كما تدير المنظمة مركزاً للإصابات البالغة في مديرية الحوبان ، وهو يعالج جرحى الحرب والمرضى الذين يحتاجون إلى الرعاية الطبية . ,
Token span (`start_token`, `end_token`)	$(2, 2)$
Character span (`start_char`, `end_char`)	$(9, 16)$
Raw types (`raw_types`)	`ORG`, `NONGOV`
Sources (`sources`)	`Wojood1_1_flat/test`; `Wojood1_1_nested/test`; `WojoodFine-Flat/split20`

Table 7: Verified TypePrediction training example.



(a) Row-normalized confusion matrix.

(b) Per-class test F_1.

(c) Macro-F_1 by sentence category.

Figure 2: TypePredictor diagnostic plots on the held-out test split.

Setting	Value
Encoder	`U4RASD/NeoAraBERT`
Classes / maximum length	21 / 512
Epochs	4
Train/eval batch	4 / 8
Gradient accumulation	4
Encoder/head LR	10^{-5} / 5×10^{-5}
Optimizer	AdamW (`adamw_torch`)
Scheduler	Linear
Weight decay / warm-up	0.01 / 0.10
Maximum gradient norm	1.0
Precision / seed	FP16 / 42
Loss	Unweighted cross-entropy
Selection metric	Validation macro-F_1

Table 8: TypePredictor training configuration.

For Qwen-RE configurations using the nearest-candidate heuristic, let the subject span be $[s_s, s_e)$ and the object span be $[o_s, o_e)$. Candidate pairs are sorted lexicographically by

1. character gap

$$d_{\text{gap}} = \max(0, \max(s_s, o_s) - \min(s_e, o_e)) ;$$

2. twice the absolute distance between span centers,

$$d_{\text{center}} = |s_s + s_e - o_s - o_e|;$$

3. ascending subject start offset s_s; and

4. ascending object start offset o_s.

The first candidate under this ordering is selected. Overlapping and nested mentions therefore receive a character gap of zero, and the final two offsets make ties deterministic.

D TypedPair Encoder Systems

D.1 Typed Directional Input

For subject and object types τ_s and τ_o, the sentence receives typed opening and closing markers such as `[SUBJ_PERS]` and `[OBJ_ORG]`. Across 21 entity types, this introduces 84 marker strings. We additionally construct the directional type vector

$$\mathbf{t}_{s,o} = [\text{onehot}(\tau_s); \text{onehot}(\tau_o)] \in \mathbb{R}^{42}, \quad (2)$$

where the first 21 coordinates always represent the subject and the final 21 represent the object.

D.2 Pair Representation and Two-Head Objective

Let $H \in \mathbb{R}^{L \times d}$ be the encoder output, and let $\mathbf{h}_c$, $\mathbf{h}_s$, and $\mathbf{h}_o$ denote the sequence-start, subject-opening, and object-opening states. The directed

Split	Rows	Accuracy	Micro-F_1	Macro-F_1	Weighted-F_1	Top-3	Top-5
Validation	12,600	97.79	97.79	95.87	97.78	99.36	99.64
Test	12,600	97.94	97.94	95.99	97.93	99.35	99.62

Table 9: TypePredictor performance. Scores are percentages.

Split	Category	Rows	Accuracy	Macro-F_1	Weighted-F_1	Top-5
Validation	Seen sentence/new span	11,661	98.04	96.33	98.04	99.70
Validation	Unseen sentence	939	94.57	86.56	94.50	98.94
Test	Seen sentence/new span	11,606	98.08	96.41	98.07	99.67
Test	Unseen sentence	994	96.28	85.06	96.18	98.99

Table 10: TypePredictor results by sentence category.

Type	Support	P	R	F_1
GPE	2,691	97.39	98.44	97.91
ORG	2,659	98.08	97.82	97.95
DATE	1,977	99.04	99.54	99.29
PERS	1,071	97.95	98.32	98.14
NORP	1,021	97.65	97.85	97.75
ORDINAL	781	98.21	98.34	98.27
OCC	770	98.70	98.96	98.83
EVENT	377	98.64	96.29	97.45
CARDINAL	372	97.79	95.16	96.46
LOC	239	96.96	93.31	95.10
WEBSITE	148	96.69	98.65	97.66
FAC	129	92.80	89.92	91.34
LAW	91	100.00	100.00	100.00
TIME	87	91.57	87.36	89.41
MONEY	42	97.37	88.10	92.50
CURR	41	97.56	97.56	97.56
LANGUAGE	33	87.50	84.85	86.15
PERCENT	31	96.43	87.10	91.53
PRODUCT	19	100.00	94.74	97.30
QUANTITY	10	90.91	100.00	95.24
UNIT	11	100.00	100.00	100.00

Table 11: Per-type TypePredictor results on the 12,600-row test split.

pair representation is

$$\mathbf{u}_{s,o} = [\mathbf{h}_c; \mathbf{h}_s; \mathbf{h}_o; |\mathbf{h}_s - \mathbf{h}_o|; \mathbf{h}_s \odot \mathbf{h}_o; \mathbf{t}_{s,o}]. \quad (3)$$

Its dimensionality is $5d + 42$. A shared MLP applies $(5d + 42) \to 1024 \to 512$ using LayerNorm, GELU, and dropout.

The existence head returns $\mathbf{z}^e \in \mathbb{R}^2$, while the positive-relation head returns $\mathbf{z}^+ \in \mathbb{R}^{40}$. For relation label y, define $a = \mathbf{1}[y \neq \textit{no_relation}]$. Training minimizes

$$\mathcal{L}_{\text{EP}} = \text{CE}(\mathbf{z}^e, a) + a\ \text{CE}(\mathbf{z}^+, y^+). \quad (4)$$

Thus, every row supervises relation existence, whereas negative rows do not assign an arbitrary target to the positive-relation head.

D.3 Row-Level Candidate Strategies

For occurrence-pair candidate c, let

$$\begin{aligned} p_c &= P_c(\text{relation exists}), \\ q_c(r) &= P_c(r \mid \text{relation exists}). \end{aligned} \quad (5)$$

after ontology-incompatible positive relations have been masked. The released pipeline contains several deterministic alternatives for combining predictions across occurrence pairs.

Majority + logit-sum. Let $P = \{c : p_c \geq \tau\}$ among N candidates. The row is treated as positive only when

$$|P| \geq \lceil 0.5N \rceil .$$

For a positive row, masked positive-relation logits are summed over the passing candidates and the relation with the largest aggregate score is selected.

Top existence. First select

$$c^* = \arg\max_c p_c.$$

If $p_{c^*} < \tau$, the row is assigned *no_relation*. Otherwise, the output is

$$\arg\max_r q_{c^*}(r).$$

Max joint. Among candidates that pass the existence threshold, select

$$(c^*, r^*) = \arg\max_{c,r} p_c q_c(r). \quad (6)$$

If no candidate passes the threshold, the result is *no_relation*.

Soft joint pooling. Positive evidence for relation r is pooled as

$$S(r) = \sum_{c: p_c \geq \tau} p_c q_c(r), \quad (7)$$

and the relation with the largest pooled score is returned when a positive decision is made. Ties are resolved deterministically using decision score, existence probability, candidate order, and relation index.

Operation	Base	Large
Encoder states	$L \times 768$	$L \times 1024$
Pair vector	3,882	5,162
Shared hidden	512	512
Existence logits	2	2
Positive logits	40	40

Table 12: TypedPair tensor dimensions.

Configuration	Official F_1
NeoAraBERT, interactions, majority + logit-sum	88.33
NeoAraBERT, top existence	88.59
AraBERTv02	89.33
NeoAraBERT, no interactions	89.36
ARBERTv2	91.96
AraBERTv02-Large, soft pool	92.20
AraBERTv02-Large, top existence	92.29
AraModernBERT	**92.69**

Table 13: Official TypedPair results.

D.4 Dimensions and Training

For $d = 768$, the MLP and heads add 4,525,610 parameters; for $d = 1024$, they add 5,836,330. The no-interactions variant uses $[\mathbf{h}_c; \mathbf{h}_s; \mathbf{h}_o; \mathbf{t}_{s,o}]$ and adds 2,952,746 parameters.

Training uses four epochs, maximum length 512, physical batch size 4, gradient accumulation 4, encoder learning rate 10^{-5}, new-layer learning rate 5×10^{-5}, weight decay 0.01, 10% warm-up, FP16, and MLP dropout 0.1. Evaluation occurs every 250 optimizer steps. For the released AraModernBERT model, the existence threshold selected on labeled validation data is 0.01, with positive micro-F_1 used for threshold selection.

D.5 Extended Official Results

Table 13 summarizes the official scores across encoder configurations, with AraModernBERT obtaining the highest reported F_1 score of 92.69.

D.6 AraModernBERT Component Ablation

AraModernBERT achieved the strongest encoder result and was therefore used to compare configurations with and without typed text markers and the directional type vector. Table 14 reports the corresponding official full-pipeline ablation submissions.

The complete configuration obtains the strongest result among these four submissions. The ablation artifacts share the intended four-epoch training setup, batch sizes, learning rates, weight decay, warm-up, MLP dimensions, dropout, and TypePredictor dependency.

Typed markers	Type vector	Official F_1
No	No	90.65
No	Yes	90.75
Yes	No	89.94
Yes	Yes	**92.69**

Table 14: AraModernBERT component ablation on the blind evaluation.

E Yehia-RE

E.1 Row-Specific Arabic Multiple Choice

Early Yehia variants generated full Arabic relation phrases and were vulnerable to explanatory text, punctuation, and harmless lexical variation. Yehia-RE instead reformulates each instance as a row-specific Arabic multiple-choice problem.

For each ontology-compatible relation, the option shown to the model is an Arabic natural-language template (`template_ar`), rather than the canonical relation identifier. Our Arabic relation templates were inspired by the relation-aware templates used in the Wojood^{Relations}/SinaTools implementation (Aljabari et al., 2025b).[1] Thus, labels such as `Affiliation.employee_of` are used internally to construct and evaluate the example, while the model sees the corresponding Arabic formulation, such as الأول يعمل لدى الثاني. The canonical relation names are not presented as answer options.

Each displayed Arabic option is assigned a verified one-token Arabic code. The available codes are أ، ب، ج، د، ه، و، ز، ح، ط، ي. The codes are row-local addresses rather than global relation-class IDs, and *no_relation* is represented by its own Arabic option لا توجد علاقة. The largest ontology-compatible candidate set contains nine positive relations, for the directed type pair PERS→ORG. Including *no_relation*, a row therefore contains at most ten options, matching the ten available Arabic answer codes.

E.2 Full-Vocabulary and Row-Local Objectives

In both variants, all prompt and padding tokens are masked from the training loss so that exactly one assistant code token is supervised.

For the full-vocabulary variant, let g be the gold code and $\mathcal{V}$ the complete model vocabulary. The

[1] https://github.com/SinaLab/sinatools

objective is

$$\mathcal{L}_{\text{FV}} = -\log \frac{\exp z_g}{\sum_{v \in \mathcal{V}} \exp z_v}. \tag{8}$$

At inference, only the codes corresponding to options displayed in the current row are compared.

The row-local variant instead normalizes directly over the displayed code set C_i:

$$\mathcal{L}_{\text{RL}} = -\log \frac{\exp z_g}{\sum_{c \in C_i} \exp z_c}. \tag{9}$$

E.3 Prompt Template

The Yehia-RE system message is:

أنت مصنّف علاقات. حدّد العلاقة الموجّهة من الكيان [الأول] إلى الكيان [الثاني].
اختر علاقة واحدة من القائمة تناسب معنى الجملة ونوعي الكيانين.
إن لم تنطبق أي علاقة فاختر «لا توجد علاقة».
أجب برمز الخيار فقط: حرف عربي واحد دون رقم أو شرح.

The row-specific user message follows this structure:

الجملة:
{marked_sentence}

نوع الأول: {first_type_ar}
نوع الثاني: {second_type_ar}

الخيارات:
أ) {option 1}
ب) {option 2}
...
{last} ({code لا توجد علاقة})

رمز الإجابة:

The assistant message contains exactly one displayed code.

E.4 Complete Training Example

The following is a fully instantiated training example from the released record `train_1831`. The canonical relation identifiers are used only internally to construct and evaluate the example; the actual prompt presents the corresponding Arabic `template_ar` options.

System message.

أنت مصنّف علاقات. حدّد العلاقة الموجّهة من الكيان [الأول] إلى الكيان [الثاني].
اختر علاقة واحدة من القائمة تناسب معنى الجملة ونوعي الكيانين.
إن لم تنطبق أي علاقة فاختر «لا توجد علاقة».
أجب برمز الخيار فقط: حرف عربي واحد دون رقم أو شرح.

User message.

الجملة:
``نجدّد نداءاتنا للسلطات التركية المعنية لتسهيل العبور العاجل للإمدادات الأساسية والموظفين إلى شمال غرب سوريا للسماح بتوسيع نطاق استجابتنا الإنسانية والطبية في المنطقة." [الأول] كريستيان ريندرز [/الأول]، منسّق أنشطة [الثاني] أطباء بلا حدود [/الثاني] في شمال غرب سوريا.

نوع الأول: شخص
نوع الثاني: منظمة

الخيارات:
أ) الأول قائد للثاني
ب) الأول مؤسس للثاني
ج) الأول رئيس للثاني
د) الأول مدير للثاني
ه) الأول عضو في الثاني
و) الأول طالب لدى الثاني
ز) الأول يعمل لدى الثاني
ح) الأول منافس للثاني
ط) الأول مالك للثاني
ي) لا توجد علاقة

رمز الإجابة:

Assistant target.

ز

The target code ز maps internally to `Affiliation.employee_of`. The canonical relation identifier is not shown to the model as an answer option; instead, the model sees its Arabic `template_ar`, الأول يعمل لدى الثاني. Only the single assistant code token contributes to the training loss.

E.5 Deterministic Transformation

For each training row, the builder validates spans and types, inserts [الأول] and [الثاني] with overlap-safe ordering, maps types to Arabic names, retrieves compatible relations, renders their Arabic `template_ar` formulations, deterministically shuffles positive options with seed 42, appends *no_relation*, assigns one-token codes, maps the gold relation to its code, and builds the chat messages.

The corresponding audit checks marker occurrence, option uniqueness, domain/range compatibility, code-token alignment, reversibility, maximum sequence length, and absence of canonical relation-identifier leakage from the answer.

E.6 QLoRA Configuration

Yehia-RE uses rank-stabilized LoRA (RS-LoRA) (Kalajdzievski, 2023). The all-linear audit finds 225 supported linear/head modules: seven internal projections in each of 32 transformer layers plus the language-model head. PEFT wraps the 224 internal modules and leaves `lm_head` frozen. Only the 448 LoRA matrices receive gradients.

E.7 One-Token Masking Audit

For each labeled row, the system message, user message, chat-template boundary, trailing template tokens, and padding positions receive label -100. Exactly one assistant code token contributes to the loss. A row is rejected if the prompt/full-conversation boundary differs, the answer code is not one token, or the sequence exceeds 1,024 tokens.

E.8 Candidate-Pair Aggregation

For occurrence-pair candidate c, let $\pi_c(k)$ denote the normalized probability of displayed option code k. Let k_c^* be the winning code, r_c its relation, and define

$$u_c = \pi_c(k_c^*)$$

as the candidate confidence. We also define the score margin

$$m_c = \ell_c^{(1)} - \ell_c^{(2)},$$

where $\ell_c^{(1)}$ and $\ell_c^{(2)}$ are the largest and second-largest option log-probabilities.

The released runner implements the following row-level strategies.

Top confidence. Select from all occurrence-pair candidates according to the lexicographic key

$$(u_c, m_c, -c),$$

where c is the deterministic candidate index. The winning candidate's relation, including *no_relation* when applicable, is returned.

Max positive. Discard candidates whose predicted relation is *no_relation*. If no positive candidate remains, return *no_relation*. Otherwise, select the remaining candidate using the same lexicographic ordering

$$(u_c, m_c, -c)$$

and return its predicted positive relation.

Majority vote. Let N be the total number of candidate occurrence pairs and let $\rho = 0.5$. Positive voting is allowed only when at least

$$\lceil \rho N \rceil$$

candidates predict a positive relation; otherwise the row is assigned *no_relation*. Positive candidates vote by predicted relation. A relation-count tie is broken by, in order:

1. the sum of normalized probability support for the tied relation over all candidates;
2. the largest winning-candidate margin for that relation; and
3. canonical relation order.

E.9 vLLM Inference

The released blind-test runner uses BF16 bitsandbytes loading, one GPU, `tensor_parallel_size=1`, maximum length 1,024, prefix caching, temperature 0, top-$p = 1$, one generated token, and an allowed-token list equal to the codes displayed in the current row. Requests are batched up to 512 prompts.

E.10 Representative Official Results

Table 16 summarizes the official results of the two Yehia-RE training variants.

Category	Setting	Value	Notes
Base	Model	`Navid-AI/Yehia-7B-preview`	pinned revision
Base	Revision	`b9dda4715eafee7e8090d2c83cfe078d75f4ebb8`	
Runtime	Quantization	4-bit NF4 + double quant.	BF16 compute
Runtime	Maximum length	1,024	gradient checkpointing
LoRA	Target	eligible internal linear layers	RS-LoRA
LoRA	r/α/dropout	16 / 32 / 0.05	39,976,960 trainable
Training	Batch/accumulation	4 / 4	effective batch 16
Training	Epochs/steps	3 / 2,943	best step 2,750
Training	LR / warm-up	5×10^{-5} / 5%	cosine, zero decay
Training	Optimizer	`paged_adamw_8bit`	clipping 1.0
Sampling	Row weight	$n_c^{-0.25}$	replacement
Evaluation	Frequency	250 steps	full development set
Seed	Data and model	42	one reported seed

Table 15: Selected Yehia-RE QLoRA configuration.

Variant	Row decision	Official F_1
Full-vocabulary	Max positive	**92.98**
Row-local	Max positive	92.84

Table 16: Representative official Yehia-RE results.

F Qwen-RE

F.1 Causal-LM Prompt Template

The Qwen-RE causal-LM prompt is constructed separately for each directed subject–object pair. It contains a system message defining the relation extraction task and required output format, followed by a user message containing the marked Arabic sentence, the subject and object strings and their predicted entity types, and only the relations compatible with the directed type pair under the operational ontology. Each compatible relation is presented with its label, Arabic template, direction note, and concise meaning. The generic prompt structure is shown below, where braced expressions denote row-specific values.

System message

You are an Arabic relation extraction system.

Given one Arabic sentence with a marked subject and object, return exactly one relation label.

You must always choose one label from the allowed label set.

Use the Arabic relation templates and direction notes as ontology definitions.

Prefer a positive relation label whenever the sentence provides plausible evidence for one.

Use *no_relation* only as a last resort when none of the allowed positive relations is supported.

Do not copy explanations, templates, or direction notes.

Return only the exact label text and nothing else.

User message

Classify the relation from the marked SUBJECT to the marked OBJECT.

The sentence uses `[SUBJ/...]`[`/SUBJ/...]` and `[OBJ/...]`[`/OBJ/...]` markers.

Use the Arabic ontology template beside each allowed label to understand its meaning.

Follow each direction note strictly because relation direction changes the label.

Choose exactly one label from the provided label set.

Prefer choosing the best-supported positive relation.

Use *no_relation* only when none of the allowed positive relations fits the marked sentence.

Task rules

1. Read the Arabic sentence and identify the relation between the marked subject and object.

2. Choose exactly one relation label.

3. Prefer the best-supported positive relation instead of defaulting to *no_relation*.

4. Use *no_relation* only when none of the allowed positive relations fits the marked sentence.

5. Use the Arabic templates and direction notes as label definitions, not as text to copy.

6. Return only the label text with no explanation.

Allowed labels and ontology meanings for this subject–object type pair

Direction: classify the relation from SUBJECT to OBJECT.

`{RelationLabel}`: Arabic template: SUBJECT OBJECT template} {relation

Direction: {relation-specific direction note}

Meaning: {concise relation meaning}

The preceding relation block is repeated for every ontology-compatible positive relation.

no_relation: Arabic template: لا توجد علاقة صالحة بين SUBJECT و OBJECT في هذه الجملة

Direction: use only when no allowed positive relation is supported by the marked sentence.

Meaning: no allowed positive relation is supported between the marked SUBJECT and OBJECT in the required direction.

Subject: {subject}

Object: {object}

Subject type: `{subject type}`

Object type: `{object type}`

Sentence:

and SUBJECT typed the containing {sentence markers} OBJECT

Answer:

During supervised fine-tuning, the assistant response following this prompt contains only the gold relation label. At inference, no gold label is provided: the constructed prompt ends at `Answer:`, after which the causal language model predicts the relation label. Because the candidate inventory is derived from the predicted directed entity-type pair, the allowed label set can differ from one input row to another.

F.2 Verified Causal-LM Training Example

The following example instantiates the prompt in Appendix F.1 using the verified training row `triple_id=3830` from Table 2.

Subject: بلدية قلقيلية

Object: قلقيلية

Subject type: `ORG`

Object type: `GPE`

Allowed relations:

`Business.has_conflict_with`: SUBJECT لديه نزاع مع OBJECT

`Location.headquartered_in`: SUBJECT يقع مقره الرئيسي في OBJECT

`Location.located_in`: SUBJECT يقع في OBJECT

`Location.nearby`: SUBJECT يقع بالقرب من OBJECT

no_relation: لا توجد علاقة صالحة بين SUBJECT و OBJECT في هذه الجملة

Sentence:

بلدية `[SUBJ/ORG]` قلقيلية `[OBJ/GPE]` قائمقام الى : `[/SUBJ/ORG]` `[/OBJ/GPE]` طولكرم / مدير منطقة قلقيلية : قرارات مجلس محلي قلقيلية : بتاريخ (2 / 1 / 1964) .

Assistant target: `Location.located_in`

During inference, the same row-specific prompt content is used without the assistant target, and the model predicts one of the allowed labels.

F.3 Generation and Complete-Label Scoring

Free-form generation can produce explanations, punctuation, partial labels, or an incompatible continuation. We therefore evaluate complete-label scoring. Let $\mathcal{C}(x)$ denote the ontology-compatible candidate set, including *no_relation*. For candidate label $c = (y_1, \ldots, y_{T_c})$, its score is the mean token log-probability

$$s(c \mid x) = \frac{1}{T_c} \sum_{i=1}^{T_c} \log p_\theta(y_i \mid x, y_{<i}). \tag{10}$$

The selected label is

$$\widehat{c} = \arg\max_{c \in \mathcal{C}(x)} s(c \mid x). \tag{11}$$

For analysis, scores can be normalized over the compatible set:

$$P(c \mid x, \mathcal{C}) = \frac{\exp s(c \mid x)}{\sum_{c' \in \mathcal{C}(x)} \exp s(c' \mid x)}. \tag{12}$$

This normalization does not change the highest-scoring label, although its value depends on the size and composition of the candidate set.

F.4 Sequence-Classifier Prompt and Objective

Qwen-RE-9B-Cls receives the same marked sentence and entity metadata but uses a shorter instruction because the model learns a fixed 41-way classification head.

A rendering for the same verified example is:

You are an expert Arabic relation extraction encoder.

Determine the semantic relation from the marked SUBJECT to the marked OBJECT. Respect relation direction and use *no_relation* only when no positive relation is stated or clearly implied. The classification head selects the final class; do not generate an answer.

Arabic sentence:

بلدية `[SUBJ/ORG]` قلقيلية `[OBJ/GPE]` قائمقام الى : `[/SUBJ/ORG]` `[/OBJ/GPE]` طولكرم / مدير منطقة قلقيلية : قرارات مجلس محلي قلقيلية : بتاريخ (2 / 1 / 1964) .

SUBJECT: بلدية قلقيلية

SUBJECT type: `ORG`

OBJECT: قلقيلية

OBJECT type: `GPE`

Property	Causal LM	Classifier
Output	Generated/scored label	One score per class
Objective	Token prediction	Multiclass CE
Inference	Generation/scoring	Masked argmax
Parsing	Required	Not required
Ontology	Prompt/candidate set	Direct logit mask

Table 17: Qwen-RE causal-LM and classifier formulations.

For input x_i, let $\mathcal{C}_i$ be the compatible relation classes, including *no_relation*. The classifier produces $\mathbf{z} \in \mathbb{R}^{41}$. Ontology-incompatible logits are masked before cross-entropy, so the effective training objective is

$$\mathcal{L}_{\text{CLS}} = -\log \frac{\exp z_{r^*}}{\sum_{r' \in \mathcal{C}_i} \exp z_{r'}}. \tag{13}$$

Rows whose gold relation is incompatible with the supplied entity types are rejected, and the same compatibility mask is used for evaluation and final prediction.

F.5 Class-Imbalance Interventions

For weighted causal-LM training, raw inverse-frequency weights are

$$w_r^{\text{raw}} = \frac{1}{n_r}. \tag{14}$$

They are normalized to mean one and clipped:

$$w_r = \min\left(5.0, \max\left(0.25, \frac{w_r^{\text{raw}}}{\frac{1}{|\mathcal{R}|}\sum_{r'} w_{r'}^{\text{raw}}}\right)\right). \tag{15}$$

Qwen-RE-9B-Cls instead uses a weighted random sampler with

$$q_r \propto (1/n_r)^{0.5}.$$

F.6 QLoRA Fine-Tuning

Qwen3.5-4B and Qwen3.5-9B are adapted with QLoRA (Dettmers et al., 2023). The frozen backbone is loaded in 4-bit NF4 with double quantization and BF16 computation. LoRA adapters target `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`.

The sequence-classification run uses train/evaluation batch sizes 4/4, gradient accumulation 4, and maximum sequence length 1,280; the recorded causal-LM evaluation configuration uses batch size 1. The classifier's recorded LoRA rank and scaling are $r = 32$ and $\alpha = 64$.

Hyperparameter	Value
Training epochs	8
Learning rate	2×10^{-4}
Scheduler / warm-up	Cosine / 0.03
Weight decay	0.0
LoRA r/α/dropout	32 / 64 / 0.05
Quantization	4-bit NF4
Double quantization	Yes
Compute dtype	bfloat16
Optimizer	Paged AdamW 8-bit
Gradient checkpointing	Yes
Evaluation interval	1,000 steps
Seed	42

Table 18: Core Qwen-RE QLoRA settings recorded by the released configurations.

Checkpoint	Positive F_1	Acc./micro-F_1
Base prompt	0.5641	0.4707
Step 1,000	0.5035	0.6129
Step 2,000	**0.7155**	**0.6977**
Step 2,943	0.7099	0.6941

Table 19: Archived Qwen-RE-9B development results.

F.7 Validation and Checkpoint Selection

Causal-LM checkpoints are evaluated on the 1,687 development examples using generation-based inference. Outputs are mapped to the relation inventory, with ontology-incompatible predictions mapped to *no_relation*, and checkpoints are selected by development micro-F_1.

The best confirmed development micro-F_1 is 0.6846 for Qwen-RE-4B and 0.6977 for Qwen-RE-9B. Table 19 shows the archived trajectory for the inspected 9B run. Since this is single-label classification, accuracy equals all-class micro-F_1; positive-only F_1 is reported separately.

Step 2,000 correctly predicts 1,177 of 1,687 examples. Because the archive does not reliably map checkpoint steps to epochs, we report steps directly. These development results are not directly comparable with the official blind scores because they use a different split and evaluation pipeline.

F.8 Extended Results

The official scores differ by 0.18 points between Qwen-RE-9B and Qwen-RE-9B-Cls.

G Qwen Exploratory Prompt Optimization

G.1 Disambiguation Rules

We manually wrote decision rules for semantically close relations using the definitions in WojoodOn-

Qwen-RE system	Official F_1
Qwen-RE-4B	91.53
Qwen-RE-9B	91.76
Qwen-RE-9B-Cls	91.94

Table 20: Official Qwen-RE results.

tology (Aljabari et al., 2025a). A representative fragment is:

> **Allowed labels and Arabic meanings:**
>
> `Location.located_in`: ordinary physical or geographic containment.
>
> `Location.headquartered_in`: OBJECT is the headquarters or principal institutional location of SUBJECT.
>
> `PartOf.geopolitical_division`: SUBJECT is an administrative or geographic division of OBJECT.
>
> **Decision rule:** Prefer the more specific headquarters, administrative-division, capital, border, or proximity relation when its evidence is present; otherwise use ordinary containment.

These rules were diagnostically useful but did not eliminate confusion. In particular, headquarters examples could still be predicted as `Location.located_in`, while longer supportive and contrastive rules reduced validation performance for the selected Qwen configuration.

G.2 GEPA Prompt Evolution

We explored GEPA, a reflective prompt-evolution method that revises prompts using model errors and validation performance (Agrawal et al., 2026). The experiment used a stratified subset of 506 training examples and 196 validation examples.

For each candidate prompt, prediction errors were summarized with the gold and predicted labels, model output, compatible labels, relation meanings, and disambiguation guidance. The reflective editor `openai/gpt-5.5` used this feedback to propose revised prompts, which were evaluated on the validation subset and selected using GEPA's Pareto-based procedure.

The best stored GEPA prompt obtained positive $F_1 = 0.6011$ and accuracy 0.5408 on the 196-example validation subset. Because the corresponding pre-GEPA score on the same subset was not recorded, we do not claim a measured improvement over the initial prompt. This experiment was exploratory and did not replace the official system.

H Additional Error Analysis

H.1 Entity-Type Errors

A hard type error can simultaneously alter the typed text markers, the 42-dimensional subject–object type vector, and the ontology-compatible relation inventory. Type prediction is therefore consequential even though TypePredictor achieves high aggregate accuracy, particularly for lower-support types and unseen-sentence mentions.

H.2 Repeated Mentions

When a surface form occurs multiple times, the system must associate the provided subject and object strings with the textual occurrences that express the relation. Nearest-candidate selection can choose an incorrect occurrence, while score-all inference increases computation and introduces competing span hypotheses. Joint occurrence and relation scoring is a natural direction for future work.

H.3 Semantically Close Relations

A recurring difficulty is distinguishing generic and specific geographic relations, particularly `Location.located_in`, `Location.headquartered_in`, and `PartOf.geopolitical_division`.
The much larger training support for `Location.located_in` can favor the generic relation when the sentence contains evidence for a more specific alternative.

H.4 Causal Generation Errors

Causal generation adds a separate failure mode: a model can assign useful semantic probability mass to the correct relation while still generating extra prose, punctuation, an alias, or only part of a canonical label. This motivated complete-label scoring in Qwen-RE and the direct 41-way classification formulation in Qwen-RE-9B-Cls.

I Reproducibility

I.1 Released Resources

The current public inventory includes:

- `U4RASD/TypePredictor`;
- `U4RASD/DRU-RE-TypedPair-TwoHead`;
- TypedPair checkpoints for AraModernBERT, AraBERTv02, ARBERTv2, AraBERTv02-Large, and NeoAraBERT-no-interactions;

- `U4RASD/DRU-RE-Yehia` and `U4RASD/DRU-RE-Yehia-2`;
- `U4RASD/WojoodRelationsAnnotated`;
- `U4RASD/Yehia-RE-SFT`;
- `U4RASD/TypePrediction`; and
- `U4RASD/DRU-RE-Qwen`.

The repository identifiers retain their released names even where the paper uses the consolidated display names Yehia-RE and Qwen-RE.

I.2 Training and Selection Summary

I.3 Software, Hardware, and Model Revisions

The released artifacts do not contain one environment manifest shared by all experiments. Table 22 therefore reports exact recorded values where available and explicitly marks information that was not preserved rather than estimating it.

TRL is not listed as a dependency of the recorded Yehia-RE training environment. Exact Python and CUDA toolkit versions for Yehia-RE, and the corresponding complete software/hardware environment for Qwen-RE, are not recoverable from the archived releases and are therefore not inferred here.

Component	Backbone	Training	Output	Selection rule
TypePredictor	NeoAraBERT	Full fine-tuning, 4 epochs	21 type logits	Validation macro-F_1
TypedPair	Arabic encoders	Full fine-tuning, 4 epochs	Existence + 40 relation logits	Development positive micro-F_1
Yehia-RE	Yehia-7B	QLoRA, 3 epochs	One Arabic code token	Development positive micro-F_1
Qwen-RE-4B/9B	Qwen3.5-4B/9B	QLoRA, 8 epochs	Canonical label text	Development micro-F_1
Qwen-RE-9B-Cls	Qwen3.5-9B	QLoRA + 41-way head	Masked class logits	Development micro-F_1

Table 21: Cross-system summary of the principal training and selection choices. Detailed configurations are reported in the corresponding system appendices.

System	Component	Recorded value	Notes
TypePredictor	Python	3.12.3	Recorded environment
TypePredictor	PyTorch	2.5.1+cu124	Recorded environment
TypePredictor	Transformers	4.49.0	Recorded environment
TypePredictor	GPU	NVIDIA A40	44.43 GiB reported
Yehia-RE	PyTorch	2.5.1	Training requirements
Yehia-RE	Transformers	4.57.1	Training requirements
Yehia-RE	PEFT	0.17.1	Training requirements
Yehia-RE	bitsandbytes	0.45.5	Training requirements
Yehia-RE	vLLM	0.10.2	Blind-test runner
Yehia-RE	GPU	NVIDIA L40S	Selected run validated on one GPU
Yehia-RE	Base revision	`b9dda471...f4ebb8`	Full hash reported in Appendix E
Qwen-RE	PyTorch	2.5.1	Training environment
Qwen-RE	Transformers	4.49.0	Training environment
Qwen-RE	PEFT	0.19.1	Recorded adapter configuration; LoRA/QLoRA adapter implementation
Qwen-RE	bitsandbytes	0.50.0	4-bit QLoRA quantization; latest PyPI release before 31 July 2026
Qwen-RE	GPU	NVIDIA RTX PRO 4500	Training hardware

Table 22: Software, hardware, and revision information recoverable from the released experiment artifacts.